

Dear Editors,

Please consider my manuscript, "Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil," for publication as a Research Paper in *Animal Behaviour*.

Understanding how selection acts on multiple, correlated traits is essential for explaining the evolution of sexual dimorphism. However, multivariate selection is rarely estimated in both sexes within the same natural population, particularly in species for which mating success can be assessed directly in the field. In this study, I examined sexual size dimorphism and phenotypic selection on body and leg length in a flightless New Zealand weevil, *Lyperobates* sp. A. I compared individuals observed in mating pairs with unpaired males and females collected from the same population.

Females were larger than males in both body and leg length, but multivariate analyses revealed broadly similar directional selection in the two sexes: mating success increased with body length but declined with leg length after accounting for their positive correlation. The male analysis additionally provided evidence of nonlinear and correlational selection, suggesting that male mating success depended on particular combinations of body and leg dimensions. Importantly, a pooled analysis provided little evidence that the directional or correlational selection gradients differed statistically between the sexes. My results therefore illustrate how similar multivariate selection can act on sexually dimorphic traits and highlight the importance of distinguishing apparent sex-specific patterns from statistically supported differences between the sexes.

I believe the manuscript is well suited to *Animal Behaviour* because it addresses the evolution of mating success, sexual dimorphism and sex-specific selection using field observations of naturally interacting animals. The study also contributes data on an endemic, flightless invertebrate and demonstrates the value of jointly analysing correlated morphological traits rather than interpreting separate univariate associations.

This manuscript is original, has not been published previously and is not under consideration elsewhere. I have disclosed funding sources in the manuscript. The research complied with applicable requirements governing the collection and treatment of wildlife in New Zealand and followed the Association for the Study of Animal Behaviour/Animal Behavior Society Guidelines for the Treatment of Animals in Behavioural Research and Teaching.

Thank you for considering my manuscript. I hope that it will be of interest to readers of *Animal Behaviour*.

Sincerely,



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